对外经济贸易大学

保险学院

教学大纲

非寿险精算学（SINS304）

授课教师： 谢远涛教授 & 黄一凡副教授

学分/课时： 3 学分 / 54 课时

授课形式： 17 次讲授 + 1 次期末答疑

授课语言： 中文（辅以英文精算术语）

学期： 2026 年秋季

1 课程基本信息

课程名称	非寿险精算学
英文名称	Casualty Actuarial Science / Non-life Actuarial Science
课程编号	SINS304 / CINS423-1
学分课时	3 学分，54 课时（讲授 48 课时 + 综合研讨 3 课时 + 期末答疑 3 课时）
课程性质	专业必修课（精算学方向）
授课对象	精算学专业高年级本科生、硕士研究生
先修课程	概率论与数理统计、保险学原理、金融数学、线性代数
授课时间	2025 年秋季学期，第 1–18 周，每周 1 次，每次 3 课时（第 18 次为期末答疑课）
授课地点	待定
授课教师	谢远涛教授、黄一凡副教授
办公时间	每周三下午 2:00–4:00（或邮件预约）
课程主页	校内在线平台 / TAS（课件、数据集、R 代码模板、作业提交）
教学资料	教材书稿 `main.tex`（第 1–11 章）、课件 `ChapX_slides.tex`、R 脚本 `ChapX.R`

2 课程概述

非寿险精算学是精算学科的核心支柱之一，涵盖财产保险、责任保险、信用保险、再保险以及各类新兴风险保险的精算建模与量化分析。本课程以非寿险精算学常用模型为主线，系统讲授损失分布、风险模型、贝叶斯与信度理论、准备金评估、再保险、时间序列分析、广义线性模型、相依风险与 Copula、极值理论、机器学习等核心内容，构建从经典精算到现代数据科学的知识体系。

课程内容基本涵盖中国精算师协会、英国精算师协会（IFoA）、北美精算师协会（SOA）、北美产险精算师协会（CAS）在准精算师阶段考试对非寿险精算部分的知识点要求，并主动对接 IFoA CS2、CAS MAS-I/MAS-II、SOA ASTAM 等考试大纲的最新变化。

本课程由谢远涛教授与黄一凡副教授共同授课，共 54 课时（17 次讲授 + 1 次期末答疑课）。教学方案同时服务于对外经济贸易大学（UIBE）精算学专业培养方案与 SOA UEC 课程认证要求，并与 SOA ASTAM、IFoA CS2、CAS MAS-I/MAS-II 等精算师考试大纲保持同步。

3 教学目标

通过本课程的学习，学生应达到以下目标：

知识目标

(1) 掌握非寿险精算的核心损失分布及其参数估计方法，理解分布拟合与优度检验的基本框架；

(2) 理解聚合风险模型与个体风险模型的构建原理，能够推导复合分布的矩和矩母函数；

(3) 掌握贝叶斯统计的基本范式，理解先验分布、后验分布、损失函数与共轭先验的概念；

(4) 理解信度理论的基本思想，掌握贝叶斯信度与经验贝叶斯信度的保费公式推导；

(5) 掌握流量三角形的构造与解读，熟练运用链梯法、案均赔款法、B-F 法进行准备金评估；

(6) 理解再保险的基本类型与定价原理，能够推导再保险下的赔付金额分布；

(7) 掌握时间序列建模的基本方法，能够识别、估计和诊断 ARIMA 模型；

(8) 理解广义线性模型的理论框架，掌握指数族分布、连接函数与参数估计方法；

(9) 理解 Copula 的基本概念与 Sklar 定理，掌握常见 Copula 函数的性质与应用；

(10) 理解极值理论的基本框架，掌握 GEV 和 GPD 模型的建模与应用；

(11) 了解机器学习的基本方法（正则回归、决策树、随机森林、神经网络等）及其在保险中的适用场景。

能力目标

(1) 能够针对具体精算问题选择合适的模型并进行参数估计与诊断；

(2) 能够使用 R 语言独立完成从数据清洗到模型拟合、诊断、解释的全流程分析；

(3) 能够阅读和理解精算学领域的英文文献和考试大纲，具备双语专业交流能力；

(4) 能够批判性地评估模型的假设、局限性和适用范围，避免机械套用；

(5) 具备将模型结果转化为业务语言（定价、准备金、资本、再保险安排）的综合能力。

思政与职业伦理目标

(1) 理解保险业作为经济减震器和社会稳定器的功能定位，增强服务国家战略的意识；

(2) 培养数据伦理意识，在建模过程中坚持实事求是、尊重数据的科学态度；

(3) 理解精算工作对社会民生的影响，树立专业责任感和职业道德；

(4) 认识人工智能与机器学习在保险定价中的公平性与透明性挑战，形成负责任的算法应用观。

4 教学内容与课时安排

周次	教学内容（对应章节与课件）	课堂活动与产出
第 1 周	**第 1 章损失分布基础（一）** 随机变量、离散分布、PGF/MGF/CGF、连续分布（Chap1_slides 1–2）	课堂推导 + R 分布作图
第 2 周	**第 1 章损失分布基础（二）** 离散分布、变换、混合、MLE、拟合优度（Chap1_slides 3–5）	fitdistrplus 拟合、优度检验报告
第 3 周	**第 2 章风险模型（一）** 短期聚合风险模型、复合泊松可加性（Chap2_slides 1）	复合分布矩与 MGF 推导
第 4 周	**第 2 章风险模型（二）** 个体风险模型、参数不确定性（Chap2_slides 2–3）	蒙特卡洛模拟、偿付能力讨论
第 5 周	**第 3 章贝叶斯与信度（一）** Bayes 定理、共轭先验、损失函数（Chap3_slides 1）	后验推导练习
第 6 周	**第 3 章贝叶斯与信度（二）** 信度公式、Bühlmann–Straub、经验贝叶斯（Chap3_slides 2–3）	信度保费计算、费率公平性案例
第 7 周	**第 4 章准备金评估（一）** 流量三角形、进展因子与链梯法（Chap4_slides 1–2）	手工计算流量三角形

周次	教学内容（对应章节与课件）	课堂活动与产出
第 8 周	**第 4 章准备金评估（二）** 案均赔款法、B-F 法、Mack/GLM 随机模型、ELR/频率-严重度/贝叶斯/泊松准备金（Chap4_slides 3–8）	ChainLadder 包实战、随机准备金模型比较
第 9 周	**第 5 章损失调整与再保险（一）** 再保险简介、再保险下的赔付金额与累积赔付金额（Chap5_slides 1–3）	再保险赔付分布推导
第 10 周	**第 5 章损失调整与再保险（二）** LER/ILF/免赔额因子/停止损失再保险、不完整数据的估计（截断、删失、通胀与 MLE）（Chap5_slides 4–7）	XOL 定价、LER/ILF 计算、停止损失定价、R 截断分布拟合
第 11 周	**第 6 章时间序列分析** 平稳性、ACF/PACF、ARMA/ARIMA、Box-Jenkins、预测、多元时间序列（Chap6_slides 1–5）	forecast 包识别、估计与预测
第 12 周	**第 7 章广义线性模型（一）** 指数族、连接函数、IRLS 与偏差（Chap7_slides 1–3）	指数族验证与对数线性模型
第 13 周	**第 7 章广义线性模型（二）** 诊断、过离散、车险费率厘定、GAM、趋势分析与定价方法对比（Chap7_slides 4–7）	glm2/glmnet 分类费率表、趋势分析、纯保费法与损失率法
第 14 周	**第 8 章相依风险与 Copula** Sklar 定理、常见 Copula、尾部相依、模拟与拟合（Chap8_slides 1–5）	copula 模拟、VaR/TVaR 比较
第 15 周	**第 9 章极值理论** GEV、POT/GPD、阈值选择、重现水平、VaR/TVaR（Chap9_slides 1–3）	extRemes 极值建模

周次	教学内容（对应章节与课件）	课堂活动与产出
第 16 周	**第 10 章机器学习（一）** 监督/无监督学习、正则回归、树、提升（`Chap10_slides` 1–3）	Lasso/RF/XGBoost 对比
第 17 周	**第 10 章机器学习（二）** 神经网络、可解释性、SHAP（`Chap10_slides` 4）	SHAP 解释、模型公平性讨论
第 18 周	**第 11 章拓展研究案例 + 期末答疑** 厚尾相依准备金、双参数提升树、贝叶斯非参数回归；重点回顾与 ASTAM 公式串讲（`Chap11_slides` 1–3）	小组展示或机考、论文复现、答疑与模拟题讲解

课时分配说明

- 本课程共 **54 课时**，采用 **17 次讲授 + 1 次期末答疑课**的形式，每次课 3 课时；
- 第 1–10 章按周次与 `ChapX_slides.tex` 课件逐节对应：第 1、2 章各占 2 周，第 3、4、5 章各占 2 周，第 6、8、9 章各占 1 周，第 7、10 章各占 2 周；
- 第 11 章拓展研究案例与期末答疑合并在第 18 周，以翻转课堂 + 集中答疑形式完成，学生需提前阅读论文并准备汇报；
- 课程设计以 SOA ASTAM 为核心主线：第 1、9 章覆盖 Severity Models，第 2 章覆盖 Aggregate Models，第 5 章覆盖 Coverage Modifications，第 1、3、7 章覆盖 Construction & Selection 与 Credibility/Pricing，第 4 章覆盖 Reserving；第 6、8、10 章为 ASTAM 外的精算前沿拓展；
- 平时成绩依托 MOOC Hourlies、R 语言作业与小组展示或机考；期末考试覆盖 ASTAM 核心与教学内容，由学校统一安排。

5 考核方式

本课程成绩同时满足 UIBE 校内要求与 SOA UEC 认证要求，具体权重如下：

考核项目	**UIBE 权重**	**UEC 权重**	**说明**
小组展示或机考 / Team Presentations or Computer-Based Exam	30%	10%	2–3 次 15–20 分钟口头汇报，含 5 分钟 Q&A；或机考形式
Hourlies / MOOC 测验	30%	10%	完成指定 MOOC 课程及其考试
期末考试（闭卷）	40%	80%	满分 100 分，提供 SOA ASTAM 公式表，仅允许 SOA 认可计算器

评分标准

- **团队展示或机考（30%）**：问题识别与表达清晰度（20%）、分析深度与工具运用（20%）、PPT 质量（20%）、展示专业度（20%）、问答表现（20%）；
- **Hourlies / MOOC（30%）**：按完成度与 MOOC 考试成绩折算；
- **期末考试（40%）**：概念理解（20%）、公式推导（40%）、计算应用（40%）；
- 所有试卷与作业均为英文；R 代码作业可用英文注释，鼓励双语关键术语对照。

6 教材与参考书目

指定教材

谢远涛，黄一凡. *非寿险精算学模型与应用*. 北京：清华大学出版社，2026.

主要参考书目

[1] Institute and Faculty of Actuaries. *Subject CS2: Actuarial Statistics 2 — Course Notes* (for the 2018/2019 exams). Actuarial Education Company (ActEd), 2018.

[2] Klugman, S.A., Panjer, H.H., & Willmot, G.E. (2019). *Loss Models: From Data to Decisions* (5th ed.). John Wiley & Sons, Inc. ISBN: 978-1-119-52375-8.

[3] Frees, E.W., Derrig, R.A., & Meyers, G. (2014). *Predictive Modeling Applications in Actuarial Science* (Vol. 1–2). Cambridge University Press.

[4] Kaas, R., Goovaerts, M., Dhaene, J., & Denuit, M. (2008). *Modern Actuarial Risk Theory: Using R* (2nd ed.). Springer.

[5] Wüthrich, M.V. & Merz, M. (2008). *Stochastic Claims Reserving Methods in Insurance.* Wiley.

[6] Coles, S. (2001). *An Introduction to Statistical Modeling of Extreme Values.* Springer.

[7] Nelsen, R.B. (2006). *An Introduction to Copulas* (2nd ed.). Springer.

[8] Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning: Data Mining, Inference, and Prediction* (2nd ed.). Springer.

[9] James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An Introduction to Statistical Learning* (2nd ed.). Springer.

[10] 中国精算师协会. *中国精算师资格考试用书——非寿险精算*. 中国财政经济出版社.

[11] Casualty Actuarial Society. *Exam MAS-I & MAS-II — Syllabus and Study Materials.*

软件工具

R 语言(版本 ≥ 4.0),推荐使用 RStudio 集成开发环境。主要使用 R 包:`fitdistrplus`、`actuar`、`ChainLadder`、`forecast`、`glm2`、`copula`、`evd`、`extRemes`、`glmnet`、`xgboost`、`randomForest`、`nnet`、`keras`。

附录：UEC ASTAM 课程覆盖情况

下表将 SOA 高级短期精算数学（ASTAM）/ UEC 课程的学习目标映射到 SINS304 的教学内容。标记”X” 的目标在课堂讲授、R 实验或考核中已完全覆盖；覆盖率超过加权课程的 95%。

编号	学习目标	权重	已覆盖
A. 损失模型 Severity Models（12–22%）			
A.1	描述参数变化如何影响分布。	2.83%	X
A.2	通过乘以常数、乘方、指数化、混合和拼接创建新分布。	2.83%	X
A.3	理解并解释这些分布的应用。	2.83%	X
A.4	基于尾部特征比较两个分布：矩、矩比、极限尾部行为、风险率和平均超额函数。	2.83%	X
A.5	理解广义极值分布和广义帕累托分布的推导和特征。	2.83%	X
A.6	将 GEV 和 GPD 应用于尾部测度和概率的估计。	2.83%	X
B. 聚合模型 Aggregate Models（12–22%）			
B.1	使用卷积和递推公式推导具有 $(a, b, 0)$ 或 $(a, b, 1)$ 频率和离散严重程度的聚合索赔分布。	5.67%	X
B.2	使用舍入法和局部矩匹配推导连续分布的离散化版本。	5.67%	X
B.3	执行复合泊松模型和的计算。	5.67%	X
C. 保障修改 Coverage Modifications（6–12%）			
C.1	评估免赔额、保单限额、最大承保损失、共同保险和停止损失保险。	3.00%	X
C.2	计算并解释损失消除比、提高限额因子和免赔额因子。	3.00%	X
C.3	评估和解释通胀对损失的影响。	3.00%	X
D. 参数模型的构建与选择 Construction and Selection of Parametric Models（12–22%）			
D.1	使用最大似然估计频率和严重度分布的参数。	3.40%	X
D.2	估计估计量的方差并构建正态和非正态置信区间。	3.40%	X
D.3	使用 delta 方法估计参数函数 MLE 的方差。	3.40%	X
D.4	使用贝叶斯估计估计参数。	3.40%	X
D.5a	使用图形程序执行模型选择。	1.13%	X
D.5b	使用 KS、卡方和似然比检验执行模型选择。	1.13%	X
D.5c	使用 SBC/BIC 和 AIC 执行模型选择。	1.13%	X
E. 信度 Credibility（12–20%）			

编号	学习目标	权重	已覆盖
E.1	解释并应用贝叶斯信度。	4.00%	X
E.2	在贝叶斯信度中应用共轭先验。	4.00%	X
E.3	应用 Bühlmann 和 Bühlmann-Straub 模型并理解它们与贝叶斯模型的关系。	4.00%	X
E.4	解释并应用非参数和半参数情况下的经验贝叶斯方法。	4.00%	X
F. 短期保险保障准备金 Reserving for Short-Term Insurance Coverages（12–20%）			
F.1a	预期损失率法。	1.60%	X
F.1b	链梯法。	1.60%	X
F.1c	Bornhuetter-Ferguson 法。	1.60%	X
F.1d	贝叶斯法。	1.60%	X
F.1e	频率和严重度法。	1.60%	X
F.2a	Mack 模型。	2.67%	X
F.2b	泊松模型。	2.67%	X
F.2c	过离散泊松模型。	2.67%	X
G. 短期保险保障定价 Pricing for Short-Term Insurance Coverages（4–12%）			
G.1	使用趋势分析计算预测损失。	2.67%	X
G.2	使用损失成本和损失率法计算整体平均费率和费率变化。	2.67%	X
G.3	计算风险分类差异变化，包括平衡回溯。	2.67%	X
总计		**100%**	**100% 完全覆盖**

说明。根据最新的课件补充，所有 UEC ASTAM 学习目标现已在课堂讲授、R 实验或考核中覆盖，包括 F.2c（过离散泊松准备金）和 G.3（分类差异变化/平衡回溯）。加权覆盖率达到 100%。

附录 B：与主教材周次安排的映射

对于 UIBE 现场授课，18 周的教学安排映射到教材《非寿险精算学模型与应用》（谢远涛、黄一凡，2026）如下。期末考试在大学考试周举行。

- **第 1 周**：第 1 章损失分布基础（一）——随机变量、离散分布、PGF/MGF/CGF。
- **第 2 周**：第 1 章损失分布基础（二）——连续分布、变换、混合、MLE、拟合优度。
- **第 3 周**：第 2 章风险模型（一）——短期聚合风险模型、复合泊松可加性。
- **第 4 周**：第 2 章风险模型（二）——个体风险模型、参数不确定性。
- **第 5 周**：第 3 章贝叶斯与信度（一）——Bayes 定理、共轭先验、损失函数。
- **第 6 周**：第 3 章贝叶斯与信度（二）——信度公式、Bühlmann–Straub、经验贝叶斯。
- **第 7 周**：第 4 章准备金评估（一）——流量三角形、进展因子与链梯法。
- **第 8 周**：第 4 章准备金评估（二）——案均赔款法、B-F 法、Mack/GLM 随机模型、ELR/频率-严重度/贝叶斯/泊松准备金。
- **第 9 周**：第 5 章损失调整与再保险（一）——再保险简介、再保险下的赔付金额与累积赔付金额。
- **第 10 周**：第 5 章损失调整与再保险（二）——LER/ILF/免赔额因子/停止损失再保险、不完整数据的估计（截断、删失、通胀与 MLE）。
- **第 11 周**：第 6 章时间序列分析——平稳性、ACF/PACF、ARMA/ARIMA、Box-Jenkins、预测、多元时间序列（ASTAM 外拓展）。
- **第 12 周**：第 7 章广义线性模型（一）——指数族、连接函数、IRLS 与偏差。
- **第 13 周**：第 7 章广义线性模型（二）——诊断、过离散、车险费率厘定、GAM、趋势分析与定价方法。
- **第 14 周**：第 8 章相依风险与 Copula——Sklar 定理、常见 Copula、尾部相依、模拟与拟合（ASTAM 外拓展）。
- **第 15 周**：第 9 章极值理论——GEV、POT/GPD、阈值选择、重现水平、VaR/TVaR。
- **第 16 周**：第 10 章机器学习（一）——监督/无监督学习、正则回归、树、提升（ASTAM 外拓展）。

- **第 17 周**：第 10 章机器学习（二）——神经网络、可解释性、SHAP（ASTAM 外拓展）。
- **第 18 周**：第 11 章拓展研究案例 + 期末答疑——厚尾相依准备金、双参数提升树、贝叶斯非参数回归；重点回顾与 ASTAM 公式串讲。

附加 UIBE 教学信息

教学方法

- **双语授课**：以中文讲授为主，专业术语和核心概念辅以英文标注；
- **理论推导与 R 语言实战并重**：每章教学分为概念讲解、公式推导、R 语言实战三个阶段；
- **案例驱动教学**：以保险行业真实问题为切入点，从具体案例引出抽象模型；
- **前后呼应，螺旋递进**：注重知识点之间的逻辑关联；
- **研讨式学习**：第 11 章拓展研究案例采用翻转课堂模式；
- **数字化与混合式教学**：利用课程主页发布预习提纲、课后阅读、R 代码模板和数据文件。

学习支持

- 建立课程微信群或在线讨论平台，及时答疑；
- 每周发布预习提纲和课后阅读材料；
- 提供 R 代码模板和数据文件，降低编程门槛；
- 第 8 周结束后安排一次阶段性反馈，调整教学节奏；
- 第 11 章研讨课前两周公布案例材料，要求学生提前阅读并准备汇报 PPT。

教师协同授课

本课程由谢远涛教授与黄一凡副教授共同设计与讲授。谢远涛教授负责课程总体框架、损失分布、风险模型、贝叶斯与信度、准备金评估、再保险等核心模块；黄一凡副教授负责时间序列分析、广义线性模型、Copula、极值理论、机器学习与拓展研究案例等统计建模与前沿模块。两位教师统一教学标准、共享课件与 R 代码，确保课程前后衔接、难度递进、案例一致。

University of International Business and Economics

School of Insurance and Economics

Syllabus

Casualty Actuarial Science (SINS304)

Instructors: Prof. Xie Yuantao & Prof. Huang Yifan

Credits / Contact Hours: 3 credits / 54 hours

Delivery: 17 lecture sessions + 1 final Q&A session

Language: English (with Chinese actuarial terminology support)

Semester: Fall 2025

Syllabus for Casualty Actuarial Science (SINS304)

COURSE CODE: SINS304

COURSE NAME: Casualty Actuarial Science

INSTRUCTORS: Prof. Xie Yuantao and Prof. Huang Yifan

CREDITS / CONTACT HOURS: 3 credits / 54 contact hours (17 sessions of lecture + 1 final Q&A session)

OVERVIEW

These are three-credit courses designed for the undergraduate students majoring in Actuarial Science, as well as Risk Management and Insurance in UIBE.

This module is one of the core modules of Actuarial Science and Actuarial Profession's examinations of for the Society of Actuaries (SOA) Advanced Short-Term Actuarial Mathematics (ASTAM) exam (US), CS2 in IOA (UK) and CAA (China).

Links to other subjects

1.This subject assumes that the student is competent with the material covered in Probability and the required knowledge for that subject in SOA (US).

This subject assumes that the student is competent with the material covered in CS1 – Actuarial Statistics 1 and the required knowledge for that subject in IOA (UK).

2.**Short-Term Actuarial Mathematics** (ASTAM) and Predictive Analytics apply the material in this subject to actuarial and statistical modelling in SOA (US).

CM1 – Actuarial Mathematics 1 and CM2 – Actuarial Mathematics 2 apply the material in this subject to actuarial and financial modelling in IOA (UK).

3.Topics in this subject are further built upon in SP1 – Health and Care Principles, SP7 – General Insurance Reserving and Capital Modelling Principles, SP8 – General Insurance Pricing Principles and SP9 – Enterprise Risk Management Principles.

ENGLISH REQUIREMENTS

All the documents and exam papers will be in English.

COURSE DESCRIPTION

The aim of the Casualty Actuarial Science subject is to provide a further grounding in statistical modelling of particular relevance to financial work, especially in non-life insurance company. It explores some further topics on advanced statistical models.

REQUIRED BOOKS AND MATERIALS

Textbooks:

1. Institute and Faculty of Actuaries, Subject CS2: Actuarial Statistics 2 Course Notes (for the 2018 exams), Actuarial Education Company (ActEd), 2018.
2. Stuart A. Klugman, Harry H. Panjer, Gordon E. Willmot, Loss Models: From Data to Decisions (Fifth Edition),John Wiley & Sons, Inc, 2019, ISBN: 978-1-119-52375-8.

Reference Textbooks:

3. Trevor Hastie, Robert Tibshirani, Jerome Friedman. The Elements of Statistical Learning Data Mining, Inference, and Prediction, Second Edition. Springer, 2009, ISBN: 978-0387848570
4. Stuart A. Klugman, Harry H. Panjer, Gordon E. Willmot, Loss Models: From Data to Decisions (Forth Edition), John Wiley & Sons, Inc, 2004, ISBN: 0471215775.
5. Stuart A. Klugman, Harry H. Panjer, Gordon E. Willmot, Loss Models: From Data to Decisions (Forth Edition),John Wiley & Sons, Inc, 2012, ISBN: 1118315324.
6. Institute and Faculty of Actuaries, Subject CS2: Actuarial Statistics 2 Core Technical Core Reading (for the 2018 exams), 2018.
7. Elisa T. Lee & John Wenyu Wang. Statistical Methods for Survival Data Analysis (Third edition), John Wiley & Sons, Inc, 2013.

Reading Textbooks:

1. Geoff Werner, Claudine Modlin, Basic Ratemaking, Casualty Actuarial Society,2009
2. Dobson A. J., An introduction to statistical modelling, Chapman & Hall, 1983, ISBN: 0412248603.
3. Hossack I. B., Pollard J. H, Zehnwirth, B., Introductory statistics with applications in general insurance, 2nd ed. Cambridge University Press, 1999, ISBN: 052165534X.

4. Daykin C. D., Pentikainen T., Pesonen, M., Practical risk theory for actuaries, Chapman & Hall, 1994, ISBN: 0412428504.

Reading Magazines:

North American Actuarial Journal、Insurance: Mathematics and Economics、Scandinavian Actuarial Journal、ASTIN Bulletin

GOALS

The aim of the Actuarial Statistics 2 subject is to provide a grounding in mathematical and statistical modelling techniques that are of particular relevance to actuarial work, including stochastic processes and survival models and their application.

Competences

On successful completion of this subject, a student will be able to:

1 describe and use statistical distributions for risk modelling.

2 describe and apply Reinsurance and Risk Models.

3 describe and apply techniques of Copula and Extreme Value Theory.

4 describe and apply basic principles of machine learning.

5 describe and apply the main concepts underlying the analysis of time series models.

6 describe and use Bayes Statistics for risk modelling.

7 describe and use Credibility Theory for modelling.

8 describe and use Empirical Bayes Credibility Theory for modelling.

9 describe and use Generalized Linear Models for pricing and reserve.

10 describe and apply techniques of Run-Off Triangles.

11 apply stochastic models for reserves, like distribution models, GLMs and Mack models.

Syllabus topics

1 Random variables and distributions for risk modelling (30-35%)

2 Time series (5-10%)

3 Bayes Statistics, Credibility Theory, Empirical Bayes Credibility Theory (30%)

4 Generalized Linear Models, Pricing and Run-Off Triangles (30-35%)

5 Machine learning (5%)

The weightings are indicative of the approximate balance of the assessment of this subject between the main syllabus topics, averaged over a number of examination sessions.

The weightings also have a correspondence with the amount of learning material underlying each syllabus topic. However, this will also reflect aspects such as:

- the relative complexity of each topic, and hence the amount of explanation and support required for it.
- the need to provide thorough foundation understanding on which to build the other objectives.
- the extent of prior knowledge which is expected.
- the degree to which each topic area is more knowledge or application based.

Skill levels

The use of a specific command verb within a syllabus objective does not indicate that this is the only form of question which can be asked on the topic covered by that objective. The Examiners may ask a question on any syllabus topic using any of the agreed command verbs, as are defined in the document Command verbs used in the Associate and Fellowship written examinations'.

Questions may be set at any skill level: Knowledge (demonstration of a detailed knowledge and understanding of the topic), Application (demonstration of an ability to apply the principles underlying the topic within a given context) and Higher Order (demonstration of an ability to perform deeper analysis and assessment of situations, including forming judgements, taking into account different points of view, comparing and contrasting situations, suggesting possible solutions and actions, and making recommendations).

The approximate split of assessment across the three skill types is 20% Knowledge, 65% Application and 15% Higher Order skills.

Grading

The grade for this course is distributed as the following table, where the **UIBE weights** are used for determining the final grade for this course at UIBE, and the **UEC weights** are used for determining the final grade for the UEC score.

Item	UIBE (weights)	UEC (weights)
Team Presentations	30%	10%
Hourlies (MOOC)	30%	10%
Final Exam	40%	80%

For UEC, the grade that appears on your transcript will be your combined grade for the full year. **Basic semester grades are determined by the final (60%), the hourlies (20%), and work in section (30%).** Small groups of section leaders grade the exams for their sections together and assign letter grades subject to approval by Teaching Assistant. This system ensures that your section leader won't have to "squeeze" your section into a particular curve, and also that your exams won't be at the mercy of a large, impersonal exam-grading machine.

Team Presentations (10%): Oral presentations consist of twice or three times 15-20 minutes presentation followed by a 5-minute question-answer session. Team members should assume the role of tutor to present analysis and answer the questions asked by other student or teacher. The topic for each team assigned by lecturer and posted in class and on TAS.

All presentations should incorporate the use of attractive, effective PowerPoint slides or other e-version slides.

Group grade on the presentation will be based on five factors:

1. The clarity and thoroughness with which team identifies and articulates the problems - 20%,

2. The depth and breadth of team's analysis and demonstrated ability to use the concepts and tools of problem analysis in a certain case - 20%,

3. The calibre of PowerPoint slides - 20%,

4. The degree of preparation, professionalism, energy, enthusiasm, and skills demonstrated in delivering part of the presentation - 20%, and

5. Team's answers to the questions posed by the class, i.e. how well team defend and support analysis and recommendations during the Q&A period - 20%.

Hourlies (massive open online courses, MOOC) (10%): Students are required to finish the MOOC course and take the exam. The final score would be weighted to the hourlies and scores:

https://www.icourse163.org/course/UIBE-1206461821?from=searchPage&outVendor=

zw_mooc_pcssjg_

This MOOC course was recorded by the instructor: Prof. Xie Yuantao.

Final Exam (80%): Final exams are out of 100 points. All exams are closed-book and will be proctored by the class instructor. The SOA ASTAM formula sheet will be provided. Only the SOA approved calculators are allowed in exams.

Videotapes of Lectures:

This course will provide a video named "An introduction to R" recorded by Professor Xie

TAS Platform: It can be found at: http://tas1.uibe.edu.cn:81/Account/Login.aspx?preurl=/Default/Default.aspx ; or it can be reached from the courses.

OUTLINES

1 Random variables and distributions for risk modelling (30-35%)

1.1 Loss distributions, with and without risk sharing

1.1.1 Describe the properties of the statistical distributions which are suitable for modelling individual and aggregate losses.

1.1.2 Explain the concepts of excesses (deductibles), and retention limits.

1.1.3 Describe the operation of simple forms of proportional and excess of loss reinsurance.

1.1.4 Derive the distribution and corresponding moments of the claim amounts paid by the insurer and the reinsurer in the presence of excesses (deductibles) and reinsurance.

1.1.5 Estimate the parameters of a failure time or loss distribution when the data is complete, or when it is incomplete, using maximum likelihood and the method of moments.

1.1.6 Use R to fit a statistical distribution to a dataset and calculate appropriate goodness of fit measures.

1.2 Compound distributions and their applications in risk modelling

1.2.1 Construct models appropriate for short term insurance contracts in terms of the numbers of claims and the amounts of individual claims.

1.2.2 Describe the major simplifying assumptions underlying the models in 1.2.1.

1.2.3 Define a compound Poisson distribution and show that the sum of independent random variables each having a compound Poisson distribution also has a compound Poisson distribution.

1.2.4 Derive the mean, variance and coefficient of skewness for compound binomial, compound Poisson and compound negative binomial random variables.

1.2.5 Repeat 1.2.4 for both the insurer and the reinsurer after the operation of simple forms of proportional and excess of loss reinsurance.

1.3 Introduction to copulas

1.3.1 Describe how a copula can be characterized as a multivariate distribution function which is a function of the marginal distribution functions of its variates, and explain how this allows the marginal distributions to be investigated separately from the dependency between them.

1.3.2 Explain the meaning of the terms dependence or concordance, upper and lower tail dependence; and state in general terms how tail dependence can be used to help select a copula suitable for modelling particular types of risk.

1.3.3 Describe the form and characteristics of the Gaussian copula and the Archimedean family of copulas.

1.4 Introduction to extreme value theory

1.4.1 Recognize extreme value distributions, suitable for modelling the distribution of severity of loss and their relationships

1.4.2 Calculate various measures of tail weight and interpret the results to compare the tail weights.

2 Time series (5-10%)

2.1 Concepts underlying time series models

2.1.1 Explain the concept and general properties of stationary, I(0), and integrated, I(1), univariate time series.

2.1.2 Explain the concept of a stationary random series.

2.1.3 Explain the concept of a filter applied to a stationary random series.

2.1.4 Know the notation for backwards shift operator, backwards difference operator, and the concept of roots of the characteristic equation of time series.

2.1.5 Explain the concepts and basic properties of autoregressive (AR), moving average (MA), autoregressive moving average (ARMA) and autoregressive integrated moving average (ARIMA) time series.

2.1.6 Explain the concept and properties of discrete random walks and random walks with normally distributed increments, both with and without drift.

2.1.7 Explain the basic concept of a multivariate autoregressive model.

2.1.8 Explain the concept of cointegrated time series.

2.1.9 Show that certain univariate time series models have the Markov property and describe how to rearrange a univariate time series model as a multivariate Markov model.

2.2 Applications of time series models

2.2.1 Outline the processes of identification, estimation and diagnosis of a time series, the criteria for choosing between models and the diagnostic tests that might be applied to the residuals of a time series after estimation.

2.2.2 Describe briefly other non-stationary, non-linear time series models.

2.2.3 Describe simple applications of a time series model, including random walk, autoregressive and cointegrated models as applied to security prices and other economic variables.

2.2.4 Develop deterministic forecasts from time series data, using simple extrapolation and moving average models, applying smoothing techniques and seasonal adjustment when appropriate.

3 Bayes Statistics, Credibility Theory, Empirical Bayes Credibility Theory (25-30%)

3.1 Bayes Statistics

3.1.1 Use Bayes' Theorem to calculate simple conditional probabilities.

3.1.2 Explain what is meant by a prior distribution, a posterior distribution and a conjugate prior distribution.

3.1.3 Derive the posterior distribution for a parameter in simple cases.

3.1.4 Explain what is meant by a loss function.

3.1.5 Use simple loss functions to derive Bayesian estimates of parameters.

3.2 Credibility Theory

3.2.1 Explain what is meant by the credibility premium formula and describe the role played by the credibility factor.

3.2.2 Explain the Bayesian approach to credibility theory and use it to derive credibility premiums in simple cases.

3.2.3 The Poisson/gamma model

3.2.4 The normal/normal model

3.3 Empirical Bayes Credibility Theory

3.3.1 Explain the empirical Bayes approach to credibility theory, in particular its similarities with and its differences from the Bayesian approach.

3.3.2 State the assumptions underlying the EBCT1.

3.3.3 Calculate credibility premiums for the EBCT2.

4 Generalized Linear Models, Pricing and Reserving (30-35%)

4.1 Generalized Linear Models and Parametric estimation

4.1.1 Be familiar with the principles of Multiple Linear Regression and the Normal Linear Model.

4.1.2 Define an exponential family of distributions. Show that the following distributions may be written in this form: binomial, Poisson, exponential, gamma, normal.

4.1.3 State the mean and variance for an exponential family, and define the variance function and the scale parameter. Derive these quantities for the distributions in 4.1.2.

4.1.4 Explain what is meant by the link function and the canonical link function, referring to the distributions in 4.1.2.

4.1.5 Explain what is meant by a variable, a factor taking categorical values and an interaction term. Define the linear predictor, illustrating its form for simple models, including polynomial models and models involving factors.

4.1.6 Define the deviance and scaled deviance and state how the parameters of a GLM may be estimated. Describe how a suitable model may be chosen by using an analysis of deviance and by examining the significance of the parameters.

4.1.7 Define the Pearson and deviance residuals and describe how they may be used.

4.1.8 Apply statistical tests to determine the acceptability of a fitted model: Pearson's Chi-square test and the Likelihood ratio test.

4.1.9 Parameter estimation using MLE method and Delta method

4.2 Pricing and Reserving

4.2.1 Define a development factor and show how a set of assumed development factors can be used to project the future development of a delay triangle.

4.2.2 Describe and apply the basic chain ladder method for completing the delay triangle.

4.2.3 Show how the basic chain ladder method can be adjusted to make explicit allowance for inflation.

4.2.4 Discuss alternative ways for deriving development factors which may be appropriate for completing the delay triangle.

4.2.5 Describe and apply the average cost per claim method for estimating outstanding claim amounts.

4.2.6 Describe and apply the Bornhuetter-Ferguson method for estimating outstanding claim amounts.

4.2.7 Describe how a statistical model can be used to underpin a run-off triangles approach.

4.2.8 Discuss the assumptions underlying the application of the methods in 4.2.1 to 4.2.7 above.

4.2.9 Apply stochastic models for reserves, like distribution models, GLMs and Mack models.

4.2.10 Show how to calculate projected losses using trend analysis

4.2.11 Show how to calculate overall average rate and rate changes using the loss cost and loss ratio method

5 Machine learning (3-5%)

5.1 Explain and apply elementary principles of machine learning

5.1.1 Explain the meaning of the term machine learning and the difference between supervised learning and unsupervised learning.

5.1.2 Explain when machine learning is an appropriate approach to problem solving and describe examples of the types of problems typically addressed by machine learning, explaining the difference between regression and classification approaches.

5.1.3 Explain in general terms the process of training, testing, and validating a machine learning algorithm.

5.1.4 Describe commonly used machine learning techniques in each of the four areas defined by the supervised/unsupervised and regression/classification splits.

5.1.5 Use R to apply neural network and decision tree techniques to simple machine learning problems.

SCHEDULES

Weeks	**content**	**Reading materials**	**Question and test**
1	Chapter 1．Loss Distributions (I) Random variables, discrete distributions, PGF/MGF/CGF (`Chap1_slides` 1–2)	Textbook: Chap.1；Textbook2: Chap.3-7,11,14,15 Reference4:Chap.4 and Chap.10, Chap.12（12.1-12.5）；	(1)Question 1.1-1.16；(2)Exam-Style question；(3)SAS,R,SPSS
2	Chapter 1．Loss Distributions (II) Continuous distributions, transformations, mixtures, MLE, goodness-of-fit (`Chap1_slides` 3–5)	Textbook: Chap.1,3；Reference4:Chap.4；	(1)Question 1.1-1.16；(2)Exam-Style question；(3)R,`fitdistrplus`
3	Chapter 2．Risk Models (I) Short-term aggregate risk model, compound Poisson additivity (`Chap2_slides` 1)	Textbook: Chap.7；Textbook2: Chap.9,12 Reference3:Chap.6；	(1)Question 3.1-3.8；(2)Exam-Style question；(3)R

Weeks	content	Reading materials	Question and test
4	Chapter 2． Risk Models (II) Individual risk model, parameter uncertainty (`Chap2_slides` 2–3)	Textbook: Chap.7； Textbook2: Chap.14	(1)Question 3.1-3.8； (2)Exam-Style question； (3)R, Monte Carlo simulation
5	Chapter 3． Bayesian and Credibility (I) Bayes' theorem, conjugate priors, loss functions (`Chap3_slides` 1)	Textbook: Chap.2； Textbook2: Chap.13 Reference4:Chap.12(12.1-12.4)；	(1)Question 2.1-2.8； (2)Exam-Style question； (3)R
6	Chapter 3． Bayesian and Credibility (II) Credibility formulas, Bühlmann–Straub, empirical Bayes (`Chap3_slides` 2–3)	Textbook: Chap.5,6； Textbook2: Chap.17,18 Reference4: Chap.16；	(1)Question 5.1-5.6； (2)Exam-Style question； (3)R
7	Chapter 4． Reserving (I) Run-off triangle, development factors and chain ladder method (`Chap4_slides` 1–2)	Textbook: Chap.11； Reference4: Chap.5-6；	(1)Question 11.1-11.10； (2)Exam-Style question； (3)R,`ChainLadder`

Weeks	content	Reading materials	Question and test
8	Chapter 4. Reserving (II) Average cost per claim, B-F method, Mack/GLM stochastic models, ELR/Frequency-Severity/Bayesian/Poisson reserving (`Chap4_slides` 3–8)	Textbook: Chap.11; Reference4: Chap.5-6;	(1)Question 11.11-11.20; (2)Exam-Style question; (3)R,`ChainLadder`
9	Chapter 5. Loss Adjustment and Reinsurance (I) Reinsurance introduction, payment distribution and cumulative payment under reinsurance (`Chap5_slides` 1–3)	Textbook: Chap.4; Textbook2: Chap.8 Reference4:Chap.5 and Chap.11;	(1)Question 2.1-2.14; (2)Exam-Style question; (3)R

Weeks	content	Reading materials	Question and test
10	Chapter 5．Loss Adjustment and Reinsurance (II) LER/ILF/deductible factors/stop-loss reinsurance, incomplete data estimation (truncation, censoring, inflation and MLE) (`Chap5_slides` 4–7)	Textbook: Chap.4；Reference4:Chap.5,11；	(1)Question 2.1-2.14；(2)Exam-Style question；(3)R, truncated distribution fitting
11	Chapter 6．Time Series Analysis Stationarity, ACF/PACF, ARMA/ARIMA, Box-Jenkins, forecasting, multivariate time series (`Chap6_slides` 1–5)	Textbook: Chap.12,13；	(1)Question 7.1-7.6；(2)Exam-Style question；(3)R,`forecast`
12	Chapter 7．GLM (I) Exponential family, link functions, IRLS and deviance (`Chap7_slides` 1–3)	Textbook: Chap.10；Reference3:Chap.3-4；Reference4:Chap.12-13；	(1)Question 10.1-10.19；(2)Exam-Style question；(3)R,`glm2`

Weeks	content	Reading materials	Question and test
13	Chapter 7．GLM (II) Diagnostics, overdispersion, auto rate making, GAM, trend analysis and pricing methods (`Chap7_slides` 4–7)	Textbook: Chap.10；Reference3:Chap.7-8；Reference4:Chap.13；	(1)Question 10.1-10.19；(2)Exam-Style question；(3)R,`glm2`,`glmnet`
14	Chapter 8．Dependent Risks and Copula Sklar's theorem, common Copulas, tail dependence, simulation and fitting (`Chap8_slides` 1–5)	Slides；Reference4:Chap.4；	(1)Question 5.1-5.2；(2)Exam-Style question；(3)R,`copula`
15	Chapter 9．Extreme Value Theory GEV, POT/GPD, threshold selection, return levels, VaR/TVaR (`Chap9_slides` 1–3)	Reference4:Chap.4；	(1)Question 6.1-6.4；(2)Exam-Style question；(3)R,`extRemes`

Weeks	content	Reading materials	Question and test
16	Chapter 10．Machine Learning (I) Supervised/unsupervised learning, regularized regression, trees, boosting (`Chap10_slides` 1–3)	Slides；Reference3:Chap.6,10；	Lasso/RF/XGBoost comparison
17	Chapter 10．Machine Learning (II) Neural networks, interpretability, SHAP (`Chap10_slides` 4)	Slides；Reference3:Chap.11-12；	SHAP explanation, model fairness
18	Chapter 11．Extended Research Cases + Final Q&A Heavy-tailed dependent reserving, two-parameter boosting trees, Bayesian nonparametric regression; key review and ASTAM formula walkthrough (`Chap11_slides` 1–3). Final examination.		

CHAPTERS

Chapter 1．Families of Distributions

1.1 Introduction

1.1.1 The insurance setting

1.1.2 The statistical background

1.2 Estimation and goodness of fit

1.2.1 Method of moments

1.2.2 Method of maximum likelihood

1.2.3 The exponential distribution

1.2.4 The Pareto distribution

1.2.5 The Weibull distribution

1.2.6 The gamma distribution

1.2.7 The lognormal distribution

1.2.8 Mixture distributions

1.2.9 Generalizations of the Pareto distribution

1.3 Reinsurance

1.3.1 Excess of loss reinsurance

1.3.2 Proportional Reinsurance

1.4 Policy excess

Aims and requirements: 1. Describe the properties of the statistical distributions which are suitable for modeling individual and aggregate losses.

2. Derive moments and moment generating functions (where defined) of loss distributions including the gamma, exponential, Pareto, generalized Pareto, normal, lognormal, Weibull and Burr distributions.

3. Apply the principles of statistical inference to select suitable loss distributions for sets of claims.

Key points and difficulties: 1. Apply the principles of statistical inference to select suitable loss distributions.

2. Estimate the parameters of a failure time or loss distribution.

3. Goodness of Fit test for different distributions

Question and test:

1. Institute and Faculty of Actuaries “Subject CT6: Statistical Methods Course Notes” Chap.3, “*Loss Models: From Data to Decisions*” Chap.4,10,12 (12.1-12.5)

2. Question：

(1)Question 3.1-3.16

(2)Exam-Style question

3. Practice：

(1)SAS,R,SPSS

Chapter 2．Reinsurance

2.1 Reinsurance arrangements

2.1.1 Excess of loss reinsurance

2.1.2 The reinsurer's conditional claims distribution

2.1.3 Proportional Reinsurance

2.2 Particular distribution

2.2.1 Lognormal distribution

2.2.2 Normal distribution

2.3 Inflation

2.4 Estimation

2.5 Policy excess

Aims and requirements: 1. Explain the concepts of excesses (deductibles), and retention limits.

2. Describe the operation of simple forms of proportional (quota share reinsurance, surplus reinsurance) and non-proportional reinsurance (excess of loss reinsurance, stop loss reinsurance).

3. Derive the distribution and corresponding moments of the claim amounts paid by the insurer and the reinsurer in the presence of excesses (deductibles) and reinsurance.

4. Estimate the parameters of a failure time or loss distribution when the data is complete, or when it is incomplete, using maximum likelihood and the method of moments.

Key points and difficulties: 1. Describe the operation of simple forms of proportional (quota share reinsurance, surplus reinsurance) and non-proportional reinsurance (excess of loss reinsurance, stop loss reinsurance).

2. Estimate the parameters of a reinsurance variable.

Question and test:

1. Institute and Faculty of Actuaries "Subject CT6: Statistical Methods Course Notes" Chap.4, "*Loss Models: From Data to Decisions*" Chap.5,11

2. Question:

(1)Question 4.1-4.14

(2)Exam-Style question

3. Practice:

(1)SAS,R

Chapter 3. Risk Models

3.1 General features of the risks covered

3.2 Models for short term insurance contracts

3.2.1 The basic model

3.2.2 Discussion of the simplifications in the basic model

3.2.3 Notation and assumptions

3.3 The collective risk model

3.3.1 The collective risk model

3.3.2 The compound Poisson distribution

3.3.3 The compound binomial distribution

3.3.4 The compound negative binomial distribution

Aims and requirements: 1. Construct models appropriate for short term insurance contracts in terms of the numbers of claims and the amounts of individual claims.

2. Describe the major simplifying assumptions underlying the models in 1.

3. Derive the moment generating function of the sum of N independent random variables; in particular when N has a binomial, Poisson, geometric or negative binomial distribution.

4. Define a compound Poisson distribution and show that the sum of independent random variables each having a compound Poisson distribution also has a compound Poisson distribution.

5. Derive the mean, variance and coefficient of skewness for compound binomial, compound Poisson and compound negative binomial random variables.

6. Derive formulae for the moment generating functions and moments of aggregate claims over a given time period for the models in 1. In terms of the corresponding functions for the distributions of claim numbers and claim amounts, stating the mathematical assumptions underlying these formulae.

Key points and difficulties: 1. Construct models appropriate for specific risk.

2. Derive formulae for the moment generating functions and moments of aggregate claims.

Question and test:

1. Institute and Faculty of Actuaries "Subject CT6: Statistical Methods Course Notes" Chap.7, "*Loss Models: From Data to Decisions*" Chap.6

2. Question:

(1)Question 7.1-7.8

(2)Exam-Style question

3. Practice:

(1)SAS,R

Chapter 4. Risk Models 2

4.1 Aggregate claim distributions under proportional and excess of loss reinsurance

4.1.1 Proportional reinsurance

4.1.2 Excess of loss reinsurance

4.2 The individual risk model

4.3 Parameter variability/uncertainty

4.3.1 Introduction

4.3.2 Variability in a heterogeneous portfolio

4.3.3 Variability in a homogeneous portfolio

4.3.4 Variability in claim numbers and claim amounts and parameter Uncertainty

Aims and requirements: 1. Derive the mean, variance and coefficient of skewness for compound binomial, compound Poisson and compound negative binomial random variables.

2. Derive formulae for the moment generating functions and moments of aggregate claims over a given time period for the models in 1. In terms of the corresponding functions for the distributions of claim numbers and claim amounts, stating the mathematical assumptions underlying these formulae.

Key points and difficulties: 1. Construct models appropriate for reinsurance risk.

2. Derive formulae for the moment generating functions and moments of reinsurance risk.

Question and test:

1. Institute and Faculty of Actuaries "Subject CT6: Statistical Methods Course Notes" Chap.8, "*Loss Models: From Data to Decisions*" Chap.14

2. Question：

(1)Question 8.1-8.6

(2)Exam-Style question

3. Practice：

(1)SAS,R

Chapter 5．Copulas

5.1 Marginal and joint distributions

5.1.1 Definition of a copula

5.2 Concordance and tail dependence

5.2.1 Concordance

5.2.2 Tail dependence

Aims and requirements: 1.Describe how a copula can be characterised as a multivariate distribution function which is a function of the marginal distribution functions of its variates, and explain how this allows the marginal distributions to be investigated separately from the dependency between them.

2.Explain the meaning of the terms dependence or concordance, upper and lower tail dependence; and state in general terms how tail dependence can be used to help select a copula suitable for modelling particular types of risk.

3.Describe the form and characteristics of the Gaussian copula and the Archimedean family of copulas.

Key points and difficulties: 1. Apply the Archimedean family of copulas.

2. Estimate the parameters of copulas.

Question and test:

1. Institute and Faculty of Actuaries “Subject CT6: Statistical Methods Course Notes” Chap.3, “*Loss Models: From Data to Decisions*” Chap.4,10,12 （12.1-12.5）

2. Question：

(1)Question 3.1-3.16

(2)Exam-Style question

3. Practice：

(1)SAS,R,SPSS

Chapter 6．Extreme value theory

6.1 Introduction

6.2 Generalised Extreme Value (GEV) distribution

6.2.1 Maximum Values

6.2.2 Generalised Extreme Value distribution

6.3 Generalised Pareto distribution

6.3.1 Threshold exceedances

6.3.2 Generalised Pareto Distribution

6.4 Measures of tail weight

6.4.1 Existence of moments

6.4.2 Limiting density ratios

6.4.3 Hazard rate

6.4.4 Mean residual life

Aims and requirements: 1. Recognize extreme value distributions, suitable for modelling the distribution of severity of loss and their relationships

2. Calculate various measures of tail weight and interpret the results to compare the tail weights.

Key points and difficulties: 1. Apply the principles of extreme value distributions.

2. Estimate the parameters of extreme value distributions.

3. Goodness of Fit test for different distributions

Question and test:

1. Institute and Faculty of Actuaries "Subject CT6: Statistical Methods Course Notes" Chap.4, "*Loss Models: From Data to Decisions*" Chap.4,10,12 (12.1-12.5)

2. Question:

(1)Question 3.1-3.16

(2)Exam-Style question

3. Practice:

(1)SAS,R,SPSS

Chapter 7. Time Series Models

7.1 Properties of a univariate time series

7.2 Stationary random series

7.2.1 Stationarity

7.3 Main linear models of time series

7.3.1 Introduction

7.3.2 The backwards shift operator and the difference operator

7.3.3 The first-order autoregressive model AR(1)

7.3.4 The autoregressive model AR(p)

7.3.5 The first-order moving average model MA(1)

7.3.6 The moving average model MA(q)

7.3.7 The autoregressive moving average process ARMA(p,q)

7.3.8 Modelling non-stationary processes: the ARIMA model

7.3.9 The Markov property

Aims and requirements: 1. Explain the concept and general properties of stationary, $I(0)$, and integrated, $I(1)$, univariate time series.

2. Explain the concept of a stationary random series.

3. Explain the concept of a filter applied to a stationary random series.

4. Know the notation for backwards shift operator, backwards difference operator, and the concept of roots of the characteristic equation of time series.

Key points and difficulties: 1. Apply the main concepts and Procedures for underlying the analysis of time series models.

2. Understand the concept of a filter applied to a stationary random series.

3. Understand the concept for stationary random series.

Question and test:

1. Institute and Faculty of Actuaries "Subject CT6: Statistical Methods Course Notes" Chap.12

2. Question：

(1)Question 12.1-12.8

(2)Exam-Style question

3. Practice：

(1)SAS,R,SPSS

Chapter 8. Time Series Models 2

8.1 Compensating for trend and seasonality

8.1.1 Detecting non-stationary series

8.1.2 Least squares trend removal

8.1.3 Differencing

8.1.4 Seasonal differencing

8.1.5 Method of Moving Averages

8.1.6 Method of Seasonal Means

8.1.7 Transformation of the data

8.2 Identification of MA(q) and AR(p) models

8.2.1 Estimation of the ACF and PACF

8.2.2 Identification of white noise

8.2.3 Identification of MA(q)

8.2.4 Identification of AR(p)

8.3 Fitting a time series model using the Box-Jenkins Methodology

8.3.1 The Box-Jenkins methodology

8.3.2 Differencing

8.3.3 Fitting an ARMA(p, q) model

8.3.4 Parameter Estimation

8.3.5 Diagnostic checking

8.4 Forecasting

8.4.1 Box-Jenkins approach to forecasting stationary time series

8.4.2 Forecasting ARIMA processes

8.4.3 Exponential smoothing

8.5 Multivariate time series models

8.5.1 Vector autoregressions

8.5.2 Cointegrated time series

8.6 Some special non-stationary and non-linear time series Models

Aims and requirements: 1. Explain the concepts and basic properties of autoregressive (AR), moving average, (MA), autoregressive moving average (ARMA) and autoregressive integrated moving average (ARIMA) time series.

2. Explain the concept and properties of discrete random walks and random walks with normally distributed increments, both with and without drift.

3. Explain the basic concept of a multivariate autoregressive model.

4. Explain the concept of cointegrated time series.

5. Show that certain univariate time series models have the Markov property and describe how to rearrange a univariate time series model as a multivariate Markov model.

6. Outline the processes of identification, estimation and diagnosis of a time series, the criteria for choosing between models and the diagnostic tests that might be applied to the residuals of a time series after estimation.

7. Describe briefly other non-stationary, non-linear time series models.

8. Describe simple applications of a time series model, including random walk, autoregressive and cointegrated models as applied to investment variables.

9. Develop deterministic forecasts from time series data, using simple extrapolation and moving average models, applying smoothing techniques and seasonal adjustment when appropriate.

Key points and difficulties: 1. Outline the processes of identification, estimation and diagnosis of a time series, the criteria for choosing between models and the diagnostic tests that might be applied to the residuals of a time series after estimation.

2. Model the stationary or non-stationary (or even non-linear) time series for a specific variable.

Question and test:

1. Institute and Faculty of Actuaries “Subject CT6: Statistical Methods Course Notes” Chap.13

2. Question：

(1)Question 13.1-13.8

(2)Exam-Style question

3. Practice：

(1)SAS,R,SPSS

Chapter 9. Machine learning

9.1 Introduction

9.2 What is Machine Learning?

9.2.1 Aims of data analysis

9.2.2 Stages of data analysis

9.2.3 Sources of data

9.2.4 The reproducibility of research

9.3 Branches of Machine Learning

9.3.1 Supervised learning

9.3.2 Unsupervised learning

9.3.3 Semi-supervised learning

9.3.4 Reinforcement learning

9.4 Concepts relevant to learning from data

9.4.1 Parameters and hyper-parameters

9.5 Examples of key supervised and unsupervised Machine Learning techniques

9.5.1 Supervised learning

9.5.2 Unsupervised learning

9.6 Applications: supervised learning

9.6.1 Penalised Generalised Linear Models

9.6.2 Naïve Bayes classification

9.6.3 Decision trees (Classification and Regression Tree algorithm)

9.7 Applications: unsupervised learning

9.7.1 K-means clustering

9.7.2 Principal components analysis

9.8 Perspectives of statisticians, data scientists and other quantitative researchers

9.8.1 Terminology

9.8.2 What is the aim of the analysis?

Aims and requirements: 1. Explain the meaning of the term machine learning and the difference between supervised learning and unsupervised learning.

2. Explain when machine learning is an appropriate approach to problem solving and describe examples of the types of problems typically addressed by machine learning, explaining the difference between regression and classification approaches.

3. Explain in general terms the process of training, testing andvalidating a machine learning algorithm.

4. Describe commonly used machine learning techniques in each of the four areas defined by the supervised/unsupervised and regression/classification splits.

5. Use R to apply neural network and decision tree techniques to simple machine learning problems.

Key points and difficulties:

1. Apply the principles of machine learning techniques in each of the four areas defined by the supervised/unsupervised and regression/classification splits.

2. Use R to apply neural network and decision tree techniques to simple machine learning problems.

Question and test:

1. Institute and Faculty of Actuaries “Subject CT6: Statistical Methods Course Notes” Chap.18

2. Question：

(1)Question 14.1-14.5

(2)Exam-Style question

3. Practice：

(1)SAS,R,SPSS

Chapter 10．Bayesian Statistics

10.1 Introduction

10.2 Bayes’ Theorem

10.2.1 An example

10.3 Prior and posterior distributions

10.3.1 Notation

10.3.2 Determining the posterior density

10.4 The loss function

10.4.1 Quadratic loss

10.4.2 Absolute error loss

10.4.3 All-or-nothing loss

10.4.4 An example

Aims and requirements: 1. Use Bayes' Theorem to calculate simple conditional probabilities.

2. Explain what is meant by a prior distribution, a posterior distribution and a conjugate prior distribution.

3. Derive the posterior distribution for a parameter in simple cases.

4. Explain what is meant by a loss function.

5. Use simple loss functions to derive Bayesian estimates of parameters.

6. Understand the mean, median and mode estimates for three type of loss function, such as Quadratic loss, absolute error loss and all-or-nothing loss.

Key points and difficulties: 1. Use Bayes' Theorem to calculate simple conditional probabilities.

2. Derive the posterior distribution for a parameter.

Question and test:

1. Institute and Faculty of Actuaries "Subject CT6: Statistical Methods Course Notes" Chap.2, "*Loss Models: From Data to Decisions*" Chap.12(12.1-12.4)

2. Question：

(1)Question 2.1-2.8

(2)Exam-Style question

3. Practice：

(1)SPSS,R

Chapter 11. Credibility Theory

11.1 Introduction

11.2 Credibility

11.2.1 The credibility premium formula

11.2.2 The credibility factor

11.3 Bayesian credibility

11.3.1 Introduction

11.3.2 The Poisson/gamma model

11.3.3 Numerical illustrations of the Poisson/gamma model

11.3.4 The normal/normal model

11.3.5 Further remarks on the normal/normal model

11.3.6 Discussion of the Bayesian approach to credibility

Aims and requirements: 1. Explain what is meant by the credibility premium formula and describe the role played by the credibility factor.

2. Explain the Bayesian approach to credibility theory and use it to derive credibility premiums in simple cases.

Key points and difficulties: 1. Explain the credibility premium formula.

2. Explain the Bayesian approach to credibility theory.

3. Introduce the credibility premium for Poisson/gamma models.

4. Introduce the credibility premium for normal/normal models.

Question and test:

1. Institute and Faculty of Actuaries “Subject CT6: Statistical Methods Course Notes” Chap.5, “*Loss Models: From Data to Decisions*” Chap.16（16.1、16.2）

2. Question：

(1)Question 5.1-5.6

(2)Exam-Style question

3. Practice：

(1)SAS,R

Chapter 12. Empirical Credibility Theory

12.1 Empirical Bayes credibility theory: Model 1

12.1.1 Introduction

12.1.2 Model 1: specification

12.1.3 Model 1: the credibility premium

12.1.4 Model 1: parameter estimation

12.1.5 Example: Credibility premium using model 1

12.2 Empirical Bayes credibility theory: Model 2

12.2.1 Introduction

12.2.2 Model 2: specification

12.2.3 Model 2: the credibility premium

12.2.4 Model 2: parameter estimation

12.2.5 Example: Credibility premium using model 2

Aims and requirements: 1. Explain the empirical Bayes approach to credibility theory, in particular its similarities with and its differences from the Bayesian approach.

2. State the assumptions underlying the two models.

3. Calculate credibility premiums for the two models.

Key points and difficulties: 1. Explain the Empirical Bayesian credibility premium formula.

2. Explain the Empirical Bayesian approach to credibility theory.

Question and test:

1. Institute and Faculty of Actuaries “Subject CT6: Statistical Methods Course Notes” Chap.6, “*Loss Models: From Data to Decisions*” Chap.16（16.4、16.5）; “*Basic Ratemaking*” Chap.12

2. Question：

(1)Question 6.1-6.5

(2)Exam-Style question

3. Practice：

(1)SAS,R

Chapter 13. Generalized Linear Models

13.1 Introduction

13.2 Exponential families

13.2.1 Normal distribution

13.2.2 Poisson distribution

13.2.3 Binomial distribution

13.2.4 Gamma distribution

13.3 Link functions and linear predictors

13.3.1 Link functions

13.3.2 Linear predictor

13.4 Parametric estimation and model selection

13.4.1 Parameter estimation using MLE

13.4.2 Delta method

13.4.3 Scaled deviance (or likelihood ratio)

13.4.4 Likelihood ratio test

13.4.5AIC and BIC metrics

13.4.7 Example

13.5 Residuals analysis and assessment of model fit

13.5.1 Acceptability of a fitted model

13.4.2 Graphical procedures

Aims and requirements: 1. Be familiar with the principles of Multiple Linear Regression and the Normal Linear Model.

2. Define an exponential family of distributions. Show that the following distributions may be written in this form: binomial, Poisson, exponential, gamma, normal.

3. State the mean and variance for an exponential family, and define the variance function and the scale parameter. Derive these quantities for the distributions in 2.

4. Explain what is meant by the link function and the canonical link function, referring to the distributions in 2.

5. Explain what is meant by a variable, a factor taking categorical values and an interaction term. Define the linear predictor, illustrating its form for simple models, including polynomial models and models involving factors.

6. Define the deviance and scaled deviance and state how the parameters of a GLM may be estimated. Describe how a suitable model may be chosen by using an analysis of deviance and by examining the significance of the parameters.

7. Define the Pearson and deviance residuals and describe how they may be used.

8. Apply statistical tests to determine the acceptability of a fitted model: Pearson's Chi-square test and the Likelihood ratio test.

Key points and difficulties: 1. Understand the principles of Generalized Linear Models.

2. Understand how to evaluate the result by statistical methods, such as the Pearson and deviance residuals.

Question and test:

1. Institute and Faculty of Actuaries "Subject CT6: Statistical Methods Course Notes" Chap.10, "*Loss Models: From Data to Decisions*" Chap.12(12.6-12.7) and Chap.13, "*Basic Ratemaking*" Chap.10

2. Question：

(1)Question 10.1-10.19

(2)Exam-Style question

3. Practice：

(1)SAS,R,SPSS

Chapter 14. Analysis of Run-Off Triangles

14.1 Introduction

14.1.1 The origins of run-off triangles

14.1.2 Presentation of claims data

14.1.3 The general statistical model

14.2 Projections using development factors

14.2.1 Run-off patterns

14.2.2 The chain ladder method

14.2.3 Model checking

14.2.4 Other methods of deriving development factors

14.2.5 Assumptions underlying the method

14.3 Adjusting for inflation

14.3.1 The inflation adjusted chain ladder method

14.4 The average cost per claim method

14.4.1 Description of method

14.4.2 Application of the method

14.4.3 Example

14.4.4 Assumptions underlying the method

14.5 Loss ratios

14.6 The Bornhuetter-Ferguson (B-F) method

14.6.1 Concept of the Bornhuetter-Ferguson method

14.6.2 Description of the method

14.6.3 Application of the method

14.6.4 Example

14.6.5 Assumptions underlying the method

14.7 Stochastic models for reserves: Distribution models, GLMs and Mack models.

14.8 Pricing with trend analysis

14.9 Pricing with loss cost and loss ration methods

14.10 Exam-style question

Aims and requirements: 1. Define a development factor and show how a set of assumed development factors can be used to project the future development of a delay triangle.

2. Describe and apply the basic chain ladder method for completing the delay triangle.

3. Show how the basic chain ladder method can be adjusted to make explicit allowance for inflation.

4. Discuss alternative ways for deriving development factors which may be appropriate for completing the delay triangle.

5. Describe and apply the average cost per claim method for estimating outstanding claim amounts.

6. Describe and apply the Bornhuetter-Ferguson method for estimating outstanding claim amounts.

7. Describe how a statistical model can be used to underpin a run-off triangles approach.

8. Discuss the assumptions underlying the application of the methods in 1. to 7. above.

9. Apply stochastic models for reserves, like distribution models, GLMs and Mack models.

Key points and difficulties: 1. Explain the concepts of development factor.

2. Understand the basic chain ladder method.

3. Understand the Bornhuetter-Ferguson method.

Question and test:

1. Institute and Faculty of Actuaries "Subject CT6: Statistical Methods Course Notes" Chap.11, "*Basic Ratemaking*" Chap.5-6

2. Question：

(1)Question 11.1-11.20

(2)Exam-Style question

3. Practice：

(1)SAS,R, Excel

Written by Yuantao Xie

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Appendix: UEC ASTAM Curriculum Coverage

The following table maps the learning objectives of the SOA Advanced Short-Term Actuarial Mathematics (ASTAM) / UEC curriculum to the contents of SINS304. Each objective marked “X” is fully covered in lectures, R labs, or assessed work; the coverage ratio exceeds 95% of the weighted curriculum.

No.	Learning objective	Weight	Covered
A. Severity Models (12–22%)			
A.1	Describe how changes in the parameters affect the distributions.	2.83%	X
A.2	Create new distributions by multiplication by a constant, raising to a power, exponentiation, mixing and splicing.	2.83%	X
A.3	Understand and interpret the applications of these distributions.	2.83%	X
A.4	Compare two distributions based on tail characteristics: moments, ratios of moments, limiting tail behavior, hazard rate, and mean excess function.	2.83%	X
A.5	Understand the derivation and characteristics of the Generalized Extreme Value and Generalized Pareto distributions.	2.83%	X
A.6	Apply the GEV and GPD to the estimation of tail measures and probabilities.	2.83%	X
B. Aggregate Models (12–22%)			
B.1	Use convolution and recursive formulas to derive the distribution of aggregate claims with $(a, b, 0)$ or $(a, b, 1)$ frequency and discrete severity.	5.67%	X
B.2	Derive the discretized version of a continuous distribution using the method of rounding and local moment matching.	5.67%	X
B.3	Perform calculations for sums of compound Poisson models.	5.67%	X
C. Coverage Modifications (6–12%)			
C.1	Evaluate deductibles, policy limits, maximum covered loss, coinsurance, and stop-loss insurance.	3.00%	X
C.2	Calculate and interpret loss elimination ratios, increased limits factors and deductible factors.	3.00%	X
C.3	Evaluate and interpret the effects of inflation on losses.	3.00%	X

No.	Learning objective	Weight	Covered
D. Construction and Selection of Parametric Models (12–22%)			
D.1	Estimate parameters for frequency and severity distributions by maximum likelihood.	3.40%	X
D.2	Estimate the variance of estimators and construct normal and non-normal confidence intervals.	3.40%	X
D.3	Use the delta method to estimate the variance of the MLE of a function of parameter(s).	3.40%	X
D.4	Estimate parameters using Bayesian estimation.	3.40%	X
D.5a	Perform model selection using graphical procedures.	1.13%	X
D.5b	Perform model selection using KS, Chi-square and likelihood-ratio tests.	1.13%	X
D.5c	Perform model selection using SBC/BIC and AIC.	1.13%	X
E. Credibility (12–20%)			
E.1	Explain and apply Bayesian credibility.	4.00%	X
E.2	Apply conjugate priors in Bayesian credibility.	4.00%	X
E.3	Apply Bühlmann and Bühlmann-Straub models and understand their relationship to Bayesian models.	4.00%	X
E.4	Explain and apply empirical Bayesian methods in nonparametric and semiparametric cases.	4.00%	X
F. Reserving for Short-Term Insurance Coverages (12–20%)			
F.1a	Expected Loss Ratio method.	1.60%	X
F.1b	Chain-Ladder method.	1.60%	X
F.1c	Bornhuetter-Ferguson method.	1.60%	X
F.1d	Bayesian method.	1.60%	X
F.1e	Frequency and Severity method.	1.60%	X
F.2a	Mack's model.	2.67%	X
F.2b	Poisson model.	2.67%	X
F.2c	Overdispersed Poisson model.	2.67%	X
G. Pricing for Short-Term Insurance Coverages (4–12%)			
G.1	Calculate projected losses using trend analysis.	2.67%	X
G.2	Calculate overall average rates and rate changes using the loss cost and loss ratio methods.	2.67%	X

No.	Learning objective	Weight	Covered
G.3	Calculate risk classification differential changes, including balancing back.	2.67%	X
TOTAL		**100%**	**100% fully covered**

Note. All ASTAM learning objectives are now fully covered in lectures, R labs, or assessed work. Recent additions include: F.2c (overdispersed Poisson reserving, covered in Chap4 slides) and G.3 (classification differential changes / balancing back, covered in Chap7 slides). Weighted coverage has reached 100%.

Additional UIBE Teaching Information

- **Course homepage / TAS:** http://tas1.uibe.edu.cn:81/Account/Login.aspx?preurl=/Default/Default.aspx
- **MOOC (hourlies):** Students are required to finish the MOOC course and take its exam. Link: https://www.icourse163.org/course/UIBE-1206461821
- **Prerequisite video:** "An introduction to R" recorded by Prof. Xie is provided on the course platform.
- **Software:** R ($\geq$ 4.0), RStudio, and SOA-approved calculators.
- **Assessment summary:**

Item	**UIBE weight**	**UEC weight**
Team Presentations	30%	10%
Hourlies (MOOC)	30%	10%
Final Exam	40%	80%

- **English requirement:** All documents and exam papers will be in English.
- **Academic integrity:** Any form of plagiarism or cheating will be reported and penalised according to UIBE regulations.